

SMART DISTANCE TRACKER

A Laser-Based Distance Measurement System

ESP8266 D1 Mini • VL53L0X V2 • KY-008 Laser Module

Introduction

- Smart Distance Tracker measures the distance between a sensor and an object without physical contact.
- The VL53L0X V2 sensor is used for distance measurement.
- ESP8266 D1 Mini controls the system and processes sensor data.
- A push button can be used to trigger or control measurements.
- The project provides a simple and fast approach to smart distance measurement.

Smart Distance Tracker

Importance of the Project

- Provides quick and contactless distance measurement.
- Reduces manual measurement work.
- Useful where physical measurement is difficult or inconvenient.
- Introduces sensors, microcontrollers and IoT concepts.
- Can be expanded into smart monitoring and automation systems.

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Components Required

Component	Qty.	Purpose
ESP8266 D1 Mini	1	Main controller and Wi-Fi connectivity
VL53L0X V2 Laser Ranging Sensor	1	Measures distance
KY-008 Laser Module	1	Laser indication
Push Button	1	Controls measurement
Breadboard	1	Circuit assembly
Jumper Wires	As required	Electrical connections
USB Cable	1	Power and programming

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Working Principle

- The VL53L0X V2 uses time-of-flight technology to measure the distance to an object.
- The sensor receives the reflected light signal from the object.
- The sensor calculates the distance and sends the data to the ESP8266.
- The ESP8266 processes the sensor reading.
- The push button can be used to trigger or control the measurement.

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Role of ESP8266 D1 Mini

- Acts as the main controller of the project.
- Receives data from the VL53L0X sensor.
- Processes the distance information.
- Provides Wi-Fi connectivity for future IoT applications.
- Can send readings to a mobile app or web dashboard after further development.

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Uses of Smart Distance Tracker

- Distance measurement
- Robotics and automation
- Smart parking systems
- Industrial object detection
- Smart home applications
- Object positioning and monitoring
- Obstacle detection
- Educational and IoT projects

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Project Benefits

- Contactless measurement
- Easy to build and operate
- Compact and portable
- Low-cost components
- Fast measurement
- Wi-Fi connectivity
- Suitable for IoT-based applications
- Can be upgraded with displays, alerts and additional sensors

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Budget – Estimated Cost

Component	Approx. Cost (₹)
ESP8266 D1 Mini	₹250
VL53L0X V2 Laser Ranging Sensor	₹350
KY-008 Laser Module	₹80
Push Button	₹10
Breadboard	₹80
Jumper Wires	₹50
USB Cable	₹80
Estimated Total	₹900

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Future Scope

- Add an OLED or LCD display to show the distance directly.
- Connect the system to a mobile application.
- Store measurements using IoT or cloud technology.
- Add voice, LED or buzzer alerts.
- Develop automatic obstacle detection.
- Use multiple sensors for improved monitoring.
- Create a Wi-Fi-based real-time distance monitoring system.

CONCLUSION

Smart Distance Tracker is a compact, low-cost and useful project for contactless distance measurement. It combines the ESP8266 D1 Mini with the VL53L0X ranging sensor and has strong potential for IoT, robotics, automation and smart monitoring applications.

THANK YOU